

Vel Tech Group of Institutions

Name: Gontu Yaswanth

Email: vtu33434@veltech.edu.in

Roll no: Vtu33434

Phone: null

Branch: VTU

Department: CSE

Batch: 2028

Degree: B.Tech - Computer Science and Engineering

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2028_NeoColab_Competitive Coding II_L8

VTU_Week 10_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 14

Section 1 : MCQ

1. Which of the following sequences of array elements forms a heap?

Answer

{23, 17, 14, 6, 13, 10, 1, 12, 7, 5}

Status : Wrong

Marks : 0/1

2. Consider a max heap, represented by the array: 23, 17, 14, 6, 13, 10, 1, 5. If a value 20 is inserted into this heap, then what will be the new max heap?

Answer

23, 20, 14, 17, 13, 10, 1, 5, 6

Status : Correct

Marks : 1/1

3. Given a binary max heap. The elements are sorted in an array as 25, 14, 16, 13, 10, 8, 12. What is the content of the array after two delete operations?

Answer

14, 13, 12, 8, 10

Status : Correct

Marks : 1/1

4. In a Min Heap, what is the condition that must be satisfied between a parent and its children?

Answer

The parent must have a lower value than its children

Status : Correct

Marks : 1/1

5. When inserting an element with a value of 8 into a Min Heap, where would it be placed if the current heap contains the elements [10, 12, 15, 20, 17]?

Answer

As the root element

Status : Correct

Marks : 1/1

6. If you're inserting an element with a value of 5 into a Min Heap, and the current heap contains the elements [11, 16, 10, 13, 18], what will the Min Heap look like after the insertion?

Answer

5, 11, 10, 13, 16, 18

Status : Correct

Marks : 1/1

7. When deleting the root element from a Min Heap, which of the following replaces it as the new root?

Answer

the last element from the tree

Status : Correct

Marks : 1/1

8. Tries are mainly used for:

Answer

All of the mentioned options

Status : Correct

Marks : 1/1

9. In a trie, how is the end of a valid word usually represented?

Answer

By using an end-of-word (terminal) marker at a node

Status : Correct

Marks : 1/1

10. Which of the following is true about tries?

Answer

All of the mentioned options

Status : Correct

Marks : 1/1

11. What is stored in a segment tree node for a Range Minimum Query (RMQ)?

Answer

Minimum element in range

Status : Correct

Marks : 1/1

12. In a Range Max Query tree, what is the internal node value?

Answer

Maximum of child values

Status : Correct

Marks : 1/1

13. Which segment tree operation is used for range max queries?

Answer

max() merge

Status : Correct

Marks : 1/1

14. What data structure is often used in segment tree nodes to count distinct elements?

Answer

Set

Status : Correct

Marks : 1/1

15. In range distinct count queries, what does each segment tree node contain?

Answer

Set of elements

Status : Correct

Marks : 1/1